INCH-POUND
MIL-DTL-32687/1
w/Amendment 1
15 May 2023
SUPERSEDING
MIL-DTL-32687/1
21 September 2021

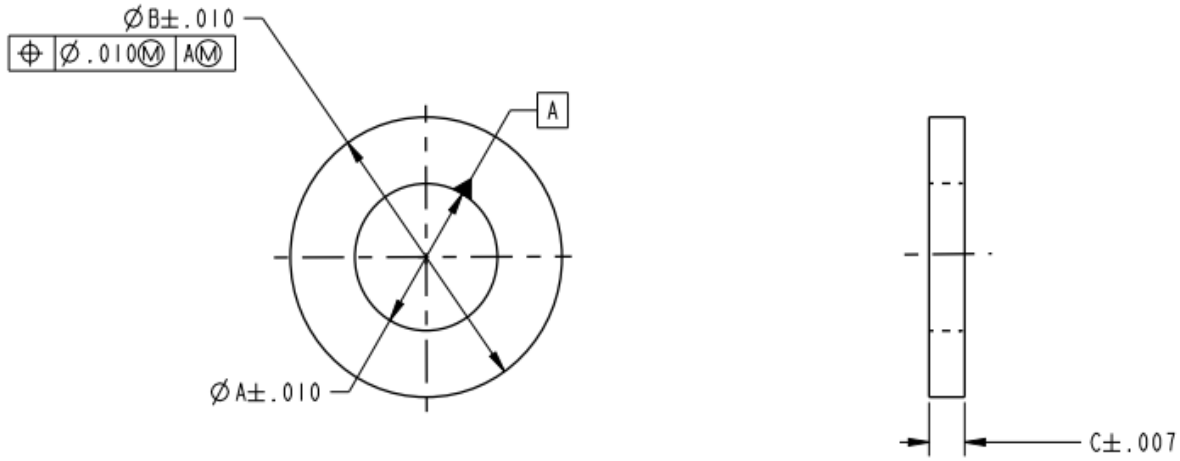
DETAIL SPECIFICATION SHEET

WASHER, FLAT, ALLOY STEEL, ROUND, HARDENED, ZINC-NICKEL PLATED

(Similar to 19207-10888410, 19207-10910174, 19207-12387327, 19207-12569613 and 19207-12662533)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32687.



NOTES:
1. Dimensions are in inches.
2. For M32687/1-50 limit deviation to specified diameter (A) is +0.010.
3. If stamping is used to manufacture washers, tolerance for dimensions A and B shall be + 0.040/ -0.010.
4. All edges and corners shall be free from burrs.
5. Unless otherwise specified, edges shall be broken R 0.02+ 0.02/ - 0.00.

FIGURE 1. Washer, flat, hardened.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5310

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REQUIREMENTS:

1 Steel composition. Unless otherwise specified, washers shall be made from one of the following materials:

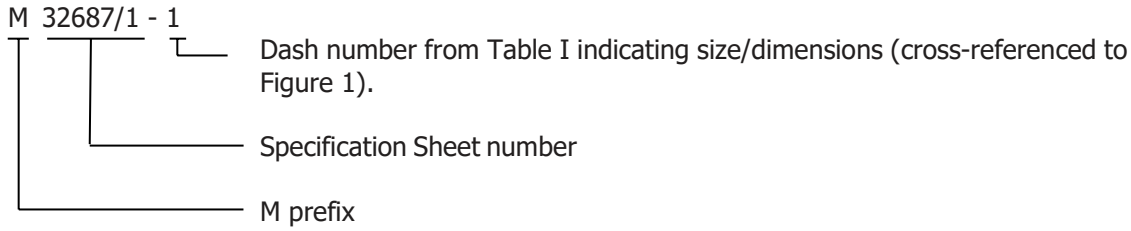
Steel, alloy, strip, sheet or plate, grade or designation 4130, 4140, or 8640, spec ASTM A506 or ASTM A829/A829M
Steel, alloy, bar, grade 4130, 4140, 8640, or 8740, spec ASTM A108 or ASTM A322
Steel, alloy, strip, sheet or plate, 4130, spec SAE AMS6350, SAE AMS6351 or SAE AMS6345

For only M32687/1-50 - Steel, carbon strip, cold-rolled, designation 1045 thru 1095, spec ASTM A684/A684M.

2 Hardness. Washers shall be quenched and tempered to HRC 33 through 38 (or equivalent hardness scale).

3 Finish. Zinc-nickel plating per MIL-PRF-32647, Type K20.

4 Part or Identifying Number (PIN). The PIN consists of the following:



Example: M32687/1-1 is a Flat Washer - Nominal 0.250 0.281 ID, 0.500 OD, and 0.089 inch thickness with Zinc-Nickel, Type K20 Plating. (See Table II for cross-reference from part numbers 19207-10888410, 19207-10910174, 19207-12387327, 19207-12569613 and 19207-12662533 to PIN.)

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TABLE I. Dimensions.

PIN	Nom Size	A	B	C		PIN	Nom Size	A	B	C
M32687/1-1	0.250	0.281	0.500	0.089		M32687/1-25	0.656	0.656	2.500	0.125
M32687/1-2	0.312	0.343	0.625	0.104		M32687/1-26	0.437	0.468	0.750	0.119
M32687/1-3	0.375	0.406	0.750	0.119		M32687/1-27	0.312	0.343	0.531	0.104
M32687/1-4	0.437	0.458	0.875	0.119		M32687/1-28	0.500	0.531	2.000	0.209
M32687/1-5	0.500	0.531	1.000	0.134		M32687/1-29	1.000	1.020	1.720	0.060
M32687/1-6	0.562	0.594	1.125	0.134		M32687/1-30	0.500	0.531	0.875	0.089
M32687/1-7	0.625	0.656	1.250	0.164		M32687/1-31	0.625	0.656	1.000	0.119
M32687/1-8	0.750	0.812	1.500	0.179		M32687/1-32	1.625	1.687	3.000	0.224
M32687/1-9	0.875	0.938	1.781	0.194		M32687/1-33	0.375	0.406	0.700	0.158
M32687/1-10	1.000	1.062	2.031	0.209		M32687/1-34	0.250	0.281	0.625	0.089
M32687/1-11	1.125	1.250	2.312	0.218		M32687/1-35	0.437	0.468	0.875	0.250
M32687/1-12	1.250	1.375	2.500	0.218		M32687/1-36	0.750	0.760	1.500	0.179
M32687/1-13	1.375	1.500	2.781	0.224		M32687/1-37	0.500	0.531	0.875	0.060
M32687/1-14	1.500	1.625	3.093	0.224		M32687/1-38	0.625	0.656	1.750	0.250
M32687/1-15	0.750	0.812	2.000	0.179		M32687/1-39	0.750	0.781	4.188	0.188
M32687/1-16	0.375	0.406	1.125	0.199		M32687/1-40	0.500	0.531	1.500	0.060
M32687/1-17	0.312	0.343	0.625	0.062		M32687/1-41	0.750	0.781	1.500	0.060
M32687/1-18	0.625	0.656	1.250	0.250		M32687/1-42	0.750	0.781	1.500	0.125
M32687/1-19	0.312	0.343	0.875	0.119		M32687/1-43	0.500	0.531	0.875	0.127
M32687/1-20	0.250	0.281	1.000	0.089		M32687/1-44	0.625	0.656	1.500	0.089
M32687/1-21	0.625	0.656	1.000	0.250		M32687/1-45	1.875	1.875	3.000	0.170
M32687/1-22	0.375	0.406	0.875	0.134		M32687/1-46	0.406	0.406	1.750	0.125
M32687/1-23	0.750	0.781	1.687	0.250		M32687/1-47	0.375	0.406	0.875	0.250
M32687/1-24	0.750	0.781	1.500	0.093		M32687/1-48	0.250	0.281	2.000	0.089
						M32687/1-50	0.750	0.812	1.500	0.125

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TABLE II. Old and new PIN. /1, /2 /3

Old Part Number	Old Part Number	Old Part Number	New Part Number
10910174-1	12387327-1	12569613-1	M32687/1-1
10910174-2	12387327-2	12569613-2	M32687/1-2
10910174-3	12387327-3	12569613-3	M32687/1-3
10910174-4	12387327-4	12569613-4	M32687/1-4
10910174-5	12387327-5	12569613-5	M32687/1-5
10910174-6	12387327-6	12569613-6	M32687/1-6
10910174-7	12387327-7	12569613-7	M32687/1-7
10910174-8	12387327-8	12569613-8	M32687/1-8
10910174-9	12387327-9	12569613-9	M32687/1-9
10910174-10	12387327-10	12569613-10	M32687/1-10
10910174-11	12387327-11	12569613-11	M32687/1-11
10910174-12	12387327-12	12569613-12	M32687/1-12
10910174-13	12387327-13	12569613-13	M32687/1-13
10910174-14	12387327-14	12569613-14	M32687/1-14
10910174-15	12387327-15	12569613-15	M32687/1-15
10910174-16	12387327-16	12569613-16	M32687/1-16
10910174-17	12387327-17	12569613-17	M32687/1-17
10910174-18	12387327-18	12569613-18	M32687/1-18
10910174-19	12387327-19	12569613-19	M32687/1-19
10910174-20	12387327-20	12569613-20	M32687/1-20
10910174-21	12387327-21	12569613-21	M32687/1-21
10910174-22	12387327-22	12569613-22	M32687/1-22
10910174-23	12387327-23	12569613-23	M32687/1-23
10910174-24	12387327-24	12569613-24	M32687/1-24

Old Part Number	Old Part Number	Old Part Number	New Part Number
10910174-25	12387327-25	12569613-25	M32687/1-25
10910174-26	12387327-26	12569613-26	M32687/1-26
10910174-27	12387327-27	12569613-27	M32687/1-27
10910174-28	12387327-28	12569613-28	M32687/1-28
10910174-29	12387327-29	12569613-29	M32687/1-29
10910174-30	12387327-30	12569613-30	M32687/1-30
10910174-31	12387327-31	12569613-31	M32687/1-31
10910174-32	12387327-32	12569613-32	M32687/1-32
10910174-33	12387327-33	12569613-33	M32687/1-33
10910174-34	12387327-34	12569613-34	M32687/1-34
10910174-35	12387327-35	12569613-35	M32687/1-35
10910174-36	12387327-36	12569613-36	M32687/1-36
10910174-37	12387327-37	12569613-37	M32687/1-37
10910174-38	12387327-38	12569613-38	M32687/1-38
10910174-39	12387327-39	12569613-39	M32687/1-39
10910174-40	12387327-40	12569613-40	M32687/1-40
10910174-41	12387327-41	12569613-41	M32687/1-41
10910174-42	12387327-42	12569613-42	M32687/1-42
10910174-43	12387327-43	12569613-43	M32687/1-43
10910174-44	12387327-44	12569613-44	M32687/1-44
10910174-45	12387327-45	12569613-45	M32687/1-45
10910174-46	12387327-46	12569613-46	M32687/1-46
	12387327-47	12662533-47	M32687/1-47
		12662533-48	M32687/1-48
		10888410	M32687/1-50

/1 The difference between the parts from the Old and New parts is the finish.
/2 Old part numbers are not to be used for new design.
/3 Use new part numbers in place of old part numbers with design authority approval.

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – AT
Navy – AS
Air Force – 70

Preparing Activity:
Army – AT

(Project 5310-2023-011)

Review Activities:
Army – AR, AV, CR4, MI
Navy – AS, OS, SH
Air Force – 11, 20

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